THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 6 — MATHS UNIT TEST SET B

Chapters 6–10: **Perimeter and Area, Fractions, Playing with Constructions, Symmetry, The Other Side of Zero** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–12 are Objective Type (1 mark each).
3. Q. 13–17 are Very Short Answer Type (2 marks each).
4. Q. 18–23 are Short Answer Type (3 marks each).
5. Q. 24–25 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 22 to 25.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 12 × 1 = 12

Choose the correct option and write it in your answer book.

1. The area of the rectangle formed by joining two squares of side 5 cm is: 1

(a) 50 cm² **(b)** 20 cm² **(c)** 100 cm² **(d)** 25 cm²

2. Area of a rectangle of length 20 cm and breadth 13 cm is: 1

(a) 260 cm² **(b)** 280 cm² **(c)** 33 cm² **(d)** 66 cm²

3. Which is greater: $\frac{3}{5}$ or $\frac{5}{8}$? 1

(a) $\frac{3}{5}$ **(b)** cannot be compared

(c) both are equal **(d)** $\frac{5}{8}$

4. In the fraction $\frac{3}{8}$, the number 8 is called the: 1

(a) denominator **(b)** numerator **(c)** remainder **(d)** quotient

5. A number made of a whole number and a proper fraction is a: 1

(a) mixed fraction **(b)** unit fraction

(c) proper fraction **(d)** improper fraction

6. To check if a drawn quadrilateral is a rectangle we can check that: 1

(a) all four angles are right angles **(b)** one side is longer

(c) its area is 10 **(d)** it has red colour

7. A square of sides 7 cm and 7 cm has perimeter: 1

(a) 49 cm **(b)** 28 cm **(c)** 14 cm **(d)** 30 cm

8. A square is a rectangle in which: 1

(a) diagonals are unequal **(b)** angles are 60°

(c) opposite sides differ **(d)** all sides are equal

9. The line along which a figure is folded to match is its: 1

(a) line of symmetry **(b)** radius **(c)** tangent **(d)** diagonal

10. Which letter has a vertical line of symmetry? 1

(a) R **(b)** F **(c)** G **(d)** A

11. $(-9) + 5$ equals: 1

(a) 14 **(b)** −14 **(c)** 4 **(d)** −4

12. The sum of an integer and its opposite is: 1

(a) the integer itself **(b)** 2

(c) 1 **(d)** 0

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 10–15 words each.

13. Find the area of a square of side 11 m. 2

14. Subtract $\frac{7}{9} - \frac{2}{9}$. 2

15. Name the instruments in a geometry box used for construction. 2

16. What is line symmetry? Give an example. 2

17. Find $9 + (-12)$. 2

SECTION — C (Short Answer Type) — 6 × 3 = 18

Answer in about 30–40 words each.

18. A rectangular garden is 25 m long and 18 m wide. Find the length of wire to fence it and its area. 3

19. A tank is $\frac{3}{4}$ full. After using $\frac{1}{8}$ of the tank, how much remains? 3

20. Does a rhombus have line symmetry? How many lines and where? 3

21. Using the token model, show $5 + (-3)$. 3

22. Subtract 3

(i) $5 - 12$
(ii) $(-3) - (-8)$
(iii) $(-6) - 4$.

OR

The temperature in Shimla was −6°C and in Jaipur 19°C. Find the difference. 3

23. The perimeter of a rectangle is 60 m and its length is 18 m. Find its breadth and area. 3

OR

How would you find the area of an irregular shape on squared paper? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 80–100 words each.

24. **Case study:** A rectangular school ground is 80 m long and 50 m wide. 5

(i) Find the length of the boundary wall.
(ii) Cost at ₹120 per metre.
(iii) Area of the ground.
(iv) Cost of levelling at ₹5 per m².

OR

(i) Derive the formula for the area of a rectangle using unit squares. 5
(ii) Find the area of a rectangle 13 m by 8 m.
(iii) Find its perimeter.

25. (i) Explain the properties of diagonals of rectangles and squares. 5
(ii) Compare them in a table.

OR

(i) State the properties of a square. 5
(ii) Construct a square of side 6 cm.
(iii) Draw its diagonals and measure the angle between them.

END OF QUESTION PAPER — Total: 50 Marks